Examining Strategies for Managing Stress and Anxiety in College Students

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Abstract

Stress and anxiety are widespread among college students, particularly in response to academic and social-evaluative stressors (Barbayannis et al., 2022). The primary purpose of this study was to investigate various strategies college students can employ to manage stress and anxiety in everyday situations. This study was designed to help better understand how students respond to a short public speaking task, specifically the speech portion of the Trier Social Stress Test (TSST), during which the participants gave a five-minute speech on why they would be a good candidate for their dream job to a panel of two confederates introduced as judges. A total of 35 undergraduate students enrolled in introductory psychology courses at Denison University participated in an in-person study assessing whether CBT-based activities, Existential therapy-based activities, or a neutral activity would reduce state anxiety immediately after completing a modified version of the TSST and after a 30-minute recovery period. Each participant was randomly assigned to one of three conditions: a CBT-based cognitive restructuring worksheet, an Existential therapy-based meaning-making worksheet, or a neutral worksheet describing a location on campus. Results from the ANOVA revealed that there were no significant differences in state anxiety between the CBT-based, Existential therapy-based, and neutral control conditions immediately following the TSST or after the 30-minute recovery period. Additionally, no condition differences were observed in the magnitude of change in anxiety from post-TSST to post-recovery. These findings do not support the hypothesis that brief CBT-based or meaning-making activities reduce state anxiety following a public speaking stressor.

Introduction

Stress is an inevitable part of the college experience, characterized by a combination of increased responsibilities, changes in daily habits, and the challenge of balancing academic, social, and personal demands. It is not uncommon for college students to experience high stress. National surveys have shown that stress is one of the most predominant mental health concerns among college students, with one study by the American College Health Association (2024) finding that 76.4% of college students reported experiencing overall moderate to high levels of stress within the past month. Other national indicators echo this pattern, with 45% of college students who seek counseling reporting stress (Winerman, 2017), and three out of four students reporting having experienced at least one stressful event in the past year (Younghans, 2018).

In particular, academic stress has been identified as a central contributor to students' mental health challenges. Specifically, one study by Barbayannis et al. (2022) found significant correlations between academic stress and poor mental well-being. Key sources of academic stress include homework overload, assessment pressure, and trying to balance academic and personal life. Such stressors produce not only psychological consequences but also physical symptoms such as insomnia and digestive issues (Pérez-Jorge et al., 2025). Additional evidence indicates that prolonged academic stress can predict academic burnout, contributing to emotional exhaustion and disengagement (Zhang et al., 2025).

Public speaking represents one of the most common and intense stressors for college students, making it a valuable context for examining stress responses. Studies estimate that 50% of college students reported having high anxiety regarding public speaking, 42% experiencing moderate anxiety, and only 9% experiencing low anxiety, highlighting how prevalent an issue it is within the college experience (Gallego et al., 2022). Public speaking anxiety is often driven by fears of being judged, fear of failure, and fear of the unknown (The Grand Valley State University Speech Lab, n.d.).

This kind of stress can be reliably reproduced through laboratory stress protocols such as the Trier Social Stress Test (TSST), which can replicate this type of evaluative anxiety through a standardized speaking task. The TSST generally consists of a speech preparation period, speech performance, and

verbal arithmetic performance periods, followed by at least one recovery period. Characterized by its social evaluation and uncontrollability, the TSST is well known for inducing a stress response that allows researchers to measure differences in reactivity, anxiety, and activation of the hypothalamic-pituitary-adrenal (HPA) or sympathetic-adrenal-medullary (SAM) axis during the task (Birkett, 2011). Studies consistently show that the TSST produces significant increases in perceived stress and anxiety during and after the task (Hellhammer & Schubert, 2012), making it a validated tool for examining short-term stress reactions and the effectiveness of stress-reduction strategies.

Because stress is so widespread among college students, identifying brief, accessible strategies to reduce anxiety is a critical priority. One well-established approach is Cognitive Behavioral Therapy (CBT), which reduces anxiety and stress by helping individuals identify, evaluate, and modify maladaptive thoughts (Curtiss et al., 2021). A core component of CBT, called cognitive restructuring, helps teach individuals how to examine evidence for and against distressing thoughts and how to replace them with more balanced and flexible interpretations (Concordia University, n.d.). According to the American Psychological Association (2022), the steps of cognitive restructuring include:

1) Write down the upsetting situation
2) Identify the most upsetting feeling you had in that situation
3) Identify your thoughts about the situation that are underlying your upsetting feelings
4) Evaluate the accuracy of your upsetting thoughts as carefully and objectively as possible
5) Decide whether your thoughts are accurate or not, based on the evidence

Numerous studies support the effectiveness of CBT-based techniques in reducing stress and anxiety in university populations. Kaya and Avci (2016) found that CBT-based skills training successfully reduced university students' trait anxiety, fear of the future, and fear of making mistakes.

Although CBT-based approaches are widely supported, recent research suggests that meaning-making, a process that is central to existential therapy, may also help individuals cope with stress. Existential therapy proposes that anxiety often arises from concerns related to uncertainty, isolation, or lack of meaning, and that individuals can reduce distress by clarifying values, strengthening a

sense of purpose, and developing greater authenticity (Center for Substance Abuse Treatment, 1999). Meaning-making involves reinterpreting stressful events in ways that align more closely with one's beliefs, values, and goals (Recovery Through Meaning Making, 2020). Empirical research increasingly supports meaning-making as an effective buffer against stress: for example, Chen et al. (2025) found that individuals adapt more positively to stress when they engage in meaning-making processes. A large meta-analysis by Vos and Vitali (2018) demonstrated that meaning-centered therapies significantly reduce psychological stress, increase meaning in life, and improve self-efficacy and optimism. These findings suggest that reflecting on one's values, sources of motivation, and personal strengths, which are core components of meaning-based interventions, may help individuals cope with stressful or evaluative situations, such as academic challenges or public speaking.

The Current Study

The present study aimed to investigate whether brief, accessible activities grounded in Cognitive Behavioral Therapy (CBT) or Existential meaning-making could reduce state anxiety in college students experiencing acute stress. Using a modified version of the Trier Social Stress Test (TSST) to elicit evaluative stress, this study compared changes in anxiety among three groups: a CBT-based cognitive reframing condition, a meaning-based existential reflection condition, and a neutral control writing condition. State anxiety was assessed at baseline, immediately after the TSST, and after a 30-minute recovery period.

Based on prior research demonstrating that both cognitive restructuring and meaning-making can reduce stress, three hypotheses were proposed. First, it was hypothesized that participants in the CBT and Existential groups would report lower STAI-6 scores immediately following the TSST compared to those in the control group. Second, it was hypothesized that participants in the CBT and Existential groups would report lower STAI-6 scores after the 30-minute recovery period relative to the control group. Finally, it was predicted that participants in the CBT and Existential groups would show a greater

reduction in STAI-6 scores from immediately post-TSST to 30 minutes post-TSST than participants in the control condition.

Methods

Design

The current study employed an in-person, between-subjects experimental design with three conditions: Cognitive Reframing (CBT-based), Meaning/Values Exploration (existential therapy-based), and a neutral control writing condition. Participants were randomly assigned to one of the three 10-minute writing conditions. State anxiety was measured at three time points: baseline, immediately after the stressor, and after the recovery period, using the 6-item State-Trait Anxiety Inventory (STAI-6) for all three time points. All participants completed the same standardized stress task, a modified version of the Trier Social Stress Test (TSST), which served as the acute stressor.

Participants

Participants were undergraduate students enrolled in psychology courses at Denison University. They were recruited through SONA, a university-managed online research participation system commonly used in psychology departments, and SignUpGenius, an online sign-up software. Participants recruited through SONA received four SONA credits, and participants recruited through SignUpGenius received four extra credit points toward their psychology course grade.

Eligibility criteria required that participants be (a) enrolled in a psychology course at Denison University and (b) be at least 18 years of age. Participation was voluntary, and informed consent was obtained prior to beginning the study.

A total of 36 undergraduate students participated in the study. Participants ranged in age from 18 to 23, with 5 aged 18, 10 aged 19, 12 aged 20, 7 aged 21, and 1 aged 23. The sample included 28% identifying as male, 68% identifying as female, and 3% identifying as another identity. Racial

composition was as follows: 16 White, 6 Black or African American, 6 Asian, and 6 identifying as another race. Academic standing included 23% freshman, 34% sophomore, 31% junior, and 11% senior.

Data were identified only by a random code consisting of the study's date, participant's initials, and the study's start time.

Materials

Each participant was randomly assigned to one of the following activities. Responses were not collected, and participants were explicitly informed that the researchers would not read any portion of their written responses.

Cognitive Reframing Worksheet

This worksheet guided participants through identifying a recent stressful situation, recognizing distressing thoughts and emotions, evaluating evidence for and against those thoughts, and creating a more balanced reframe. Based on Cognitive Behavioral Therapy (CBT), this activity emphasizes modifying maladaptive thinking patterns to reduce psychological distress. Previous research has supported that the use of brief cognitive restructuring exercises can lower perceived stress and improve emotional regulation (Kaya & Avci, 2016). Participants are asked to:

1) Describe a recent stressful academic event (e.g., exam, assignment, presentation);
2) Identify the distressing thought most central to the event;
3) Provide evidence supporting and contradicting the thought;
4) Formulate a more balanced interpretation of the event; and
5) Reflect on how they could apply this technique in future stressful situations.

The expectation was that participants would use this reframing strategy to reduce anxiety prior to the speaking task.

Meaning/Values Exploration Worksheet

This worksheet focuses on exploring participants' core values, motivations, strengths, and sources of meaning by encouraging reflection on resilience and growth rather than solutions-based or problem-solving approaches to reducing stress. The rationale for this worksheet is grounded in existential psychology, which emphasizes the role of meaning-making and self-understanding in coping with adversity (Heidenreich et al., 2021). Similar to the CBT group, participants in this group were asked to identify their motivations and strengths that allow them to overcome challenges that they may encounter in college, and how they have grown as a person over time. Finally, participants brainstormed what it would look like for them to intentionally apply and consider these strengths and motivations to future challenges. For this group, the hope was that participants would use these skills to build confidence in their ability to overcome challenges, specifically the speaking task in the study.

Neutral Control Worksheet

The control worksheet involved writing about a familiar campus location, Slayter Union, and describing it in as much detail as possible. This activity controlled for the general effects of writing and reflection without activating cognitive restructuring or self-exploration mechanisms. It ensured that differences in reported anxiety between the experimental groups and the control group are due to the activities (worksheets) themselves rather than the effects of a 10-minute-long independent writing prompt.

Stress Induction Task

A modified version of the Trier Social Stress Test (TSST) was used, consisting only of the public speaking portion. Participants were given 5 minutes to prepare for a 5-minute speech explaining why they would be a strong candidate for their dream job. The speech was delivered to a panel of two trained confederates who maintained neutral expressions throughout. Conferederates consisted of the researchers and classmates trained to adhere to the TSST evaluation procedures.

Recovery Tasks

To allow stress levels to return to baseline while keeping participants mildly engaged naturally, all individuals completed a 30-minute sequence of neutral tasks:

1. Category-Listing Task (10 minutes): writing down as many words as they can that fall under a certain category (e.g., fruits, animals).
2. Letter Cancellation Task: crossing out all instances of the letter "A" in a three-page sequence of random letters.
3. Bead-Sorting Task: sorting a specified bead color from a bowl of mixed beads using tweezers for the remainder of the recovery period.

Questionnaires

State anxiety was assessed using the 6-item State-Trait Anxiety Inventory (STAI-6), administered at three time points: baseline, immediately after the TSST, and after the 30-minute recovery period. The STAI-6 is a brief version of the original 20-item State Anxiety scale and is commonly used in studies requiring repeated measurement due to its efficiency and low participant burden.

The 6-item scale has strong empirical support for its reliability and validity. Tluczek et al. (2009) demonstrated that the specific item combination used in this study (items 1, 3, 6, 15, 16, and 17 from the full STAI-State form) showed excellent internal consistency across multiple time points ($\alpha > 0.90$), a better model fit than other short-form variations, and very high correlations with the full state anxiety scale. Similarly, Court et al. (2010), using Rasch analysis, confirmed that these six items function cohesively as a valid, unidimensional interval scale for assessing state anxiety. Together, these findings support the STAI-6 as a psychometrically sound instrument well-suited for research evaluating short-term changes in stress.

Procedure

Upon arrival, participants were greeted individually and provided with a verbal description of the study procedures, risks, and their rights. Participants then reviewed and electronically signed an informed consent form in Qualtrics.

After consent, participants completed the first STAI-6 (baseline). They were then given one of the three randomly assigned worksheets and instructed to work on it independently for 10 minutes. Researchers reiterated that responses would remain private and would not be collected.

Participants next completed the speaking portion of the TSST. They were given 5 minutes of preparation time and a notepad to take notes, but were told that they weren't allowed to bring in the notes. After the preparation period, they were taken to the room where the panelists were located, and where they would deliver their speech. Confederates were instructed to maintain flat affect and provide no feedback to standardize the stressor.

Immediately after the TSST, participants were taken back to the computer room and completed the second STAI-6. They then began the 30-minute recovery period, which consisted of the category listing task, letter cancellation task, and the bead sorting task.

After 30 minutes, participants completed the final STAI-6 and were then shown a digital copy of the debriefing form. The researcher then conducted a thorough verbal debriefing, explaining the purpose of the study and the use of deception (confederate panelists), and provided a printed copy of the debrief form. Participants were allowed to ask questions before being dismissed. The entirety of the study took approximately one hour to complete. Upon completion of the study, participants were rewarded with four SONA credits or four extra credit points. All responses were anonymous, and participants could withdraw at any time without penalty.

Data Analysis

All analyses were conducted using IBM SPSS statistical software.

Results

Overview of Hypotheses

The study tested three hypotheses related to students' anxiety levels pre-TSST, post-TSST, and after the recovery period.

1. H1: The CBT and Existential groups will report lower STAI-6 scores immediately after the TSST.
2. H2: The CBT and Existential groups will report lower STAI-6 scores after the 30-minute recovery period.
3. H3: The CBT and Existential groups will show a greater reduction in STAI-6 scores from immediately after the TSST to after the recovery period.

Descriptive Statistics

Table 1 displays the mean and standard deviation for STAI-6 scores at each survey point in the study: baseline, post-TSST, and post-recovery.

Table 1

Descriptive Statistics for STAI-6 Scores

Time	Mean STAI-6 Scores
1 (baseline)	13.11
2 (post-TSST)	14.29
3 (post-recovery)	10.66

The table illustrates how the TSST effectively increased state anxiety levels, and how anxiety levels also decreased significantly after the recovery period.

STAI-6 Scores Immediately Post-TSST

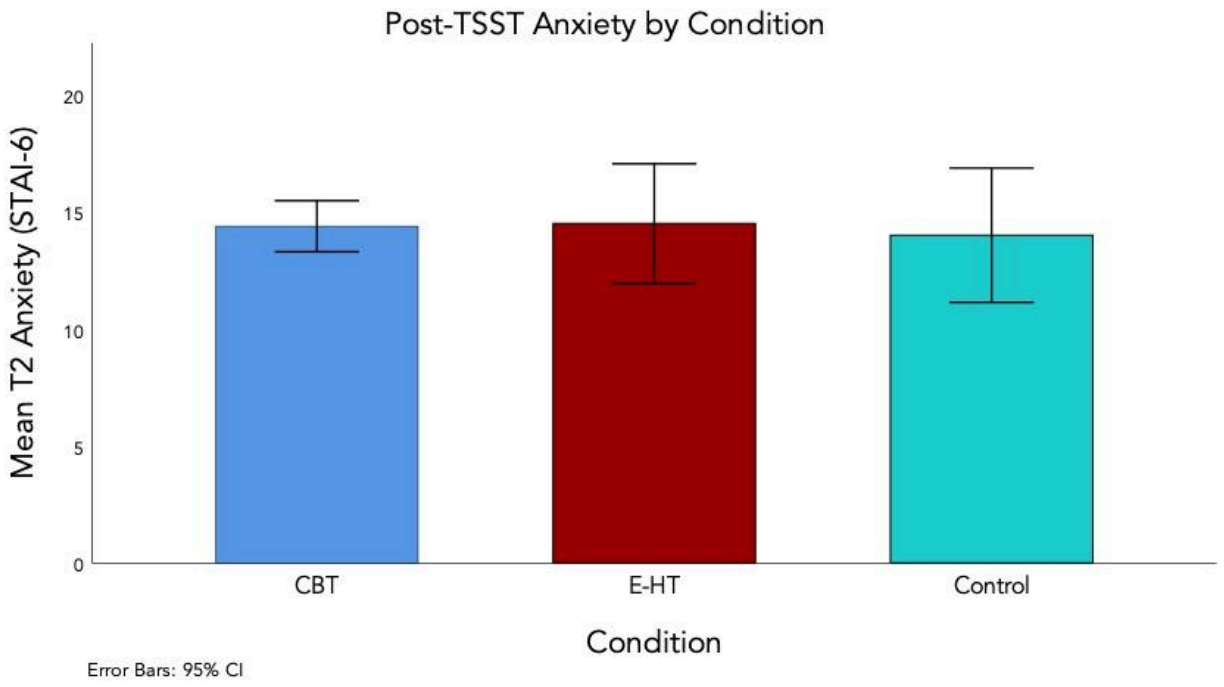
To test the first hypothesis, a one-way between-subjects ANOVA was conducted to examine whether state anxiety immediately following the TSST differed by condition (CBT, Existential, neutral

control). Results indicated that there was no significant effect of condition on post-TSST STAI-6 scores, $F(2, 33) = 0.07$, $p = 0.94$. The mean square between groups was 0.78.

Thus, hypothesis 1 was not supported, as participants in the CBT and Existential conditions did not report significantly lower state anxiety immediately after the TSST compared to participants in the control condition. A graph depicting the results can be seen in Figure 1, clearly showing that there is minimal difference in STAI-6 scores between the different condition groups.

Figure 1

Post-TSST STAI-6 Scores by Condition



STAI-Scores Post-Recovery

To test the second hypothesis, a one-way between-subjects ANOVA was conducted to examine whether state anxiety after the 30-minute recovery period differed by condition. Results indicated that there was no significant effect of condition on post-recovery STAI-6 scores, $F(2, 33) = 0.40$, $p = 0.68$.

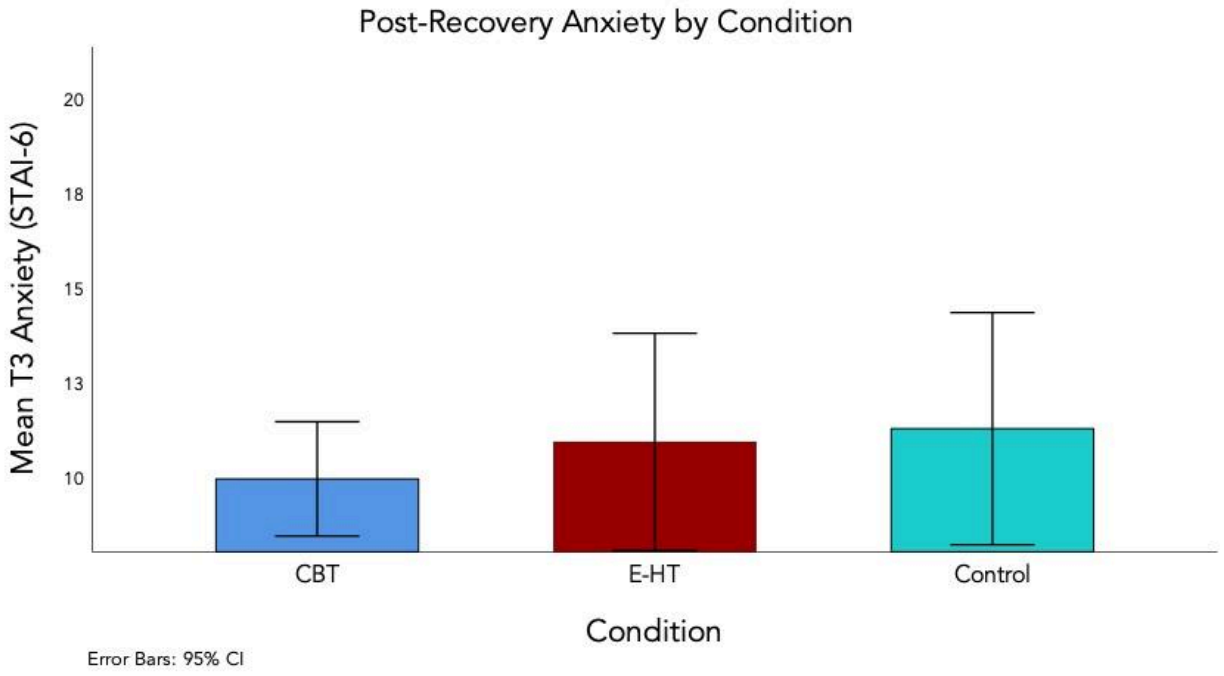
Post hoc multiple comparisons revealed that none of the pairwise differences between conditions were statistically significant. Specifically, post-recovery STAI-6 scores in the CBT condition did not

differ significantly from those in the Existential conditions (mean difference = -0.98, p = 0.82), nor from the control condition (mean difference = -1.33, p = 0.669). Additionally, STAI-6 scores in the Existential condition did not differ significantly from those in the control condition (mean difference = 0.35, p = 0.98).

Thus, hypothesis 2 was not supported, as participants in the CBT and Existential conditions did not report significantly lower anxiety after the recovery period compared to participants in the control condition (See Figure 2).

Figure 2

Post-Recovery STAI-6 Scores by Condition



Change in STAI-6 Scores from Post-TSST to Post-Recovery

To test the third hypothesis, a one-way between-subjects ANOVA was conducted to examine whether the change in state anxiety from immediately after the TSST to after the 30-minute recovery period differed by condition. Change scores were calculated by subtracting post-TSST STAI-6 scores

from post-recovery STAI-6 scores, such that positive values reflected an increase in anxiety over the recovery period.

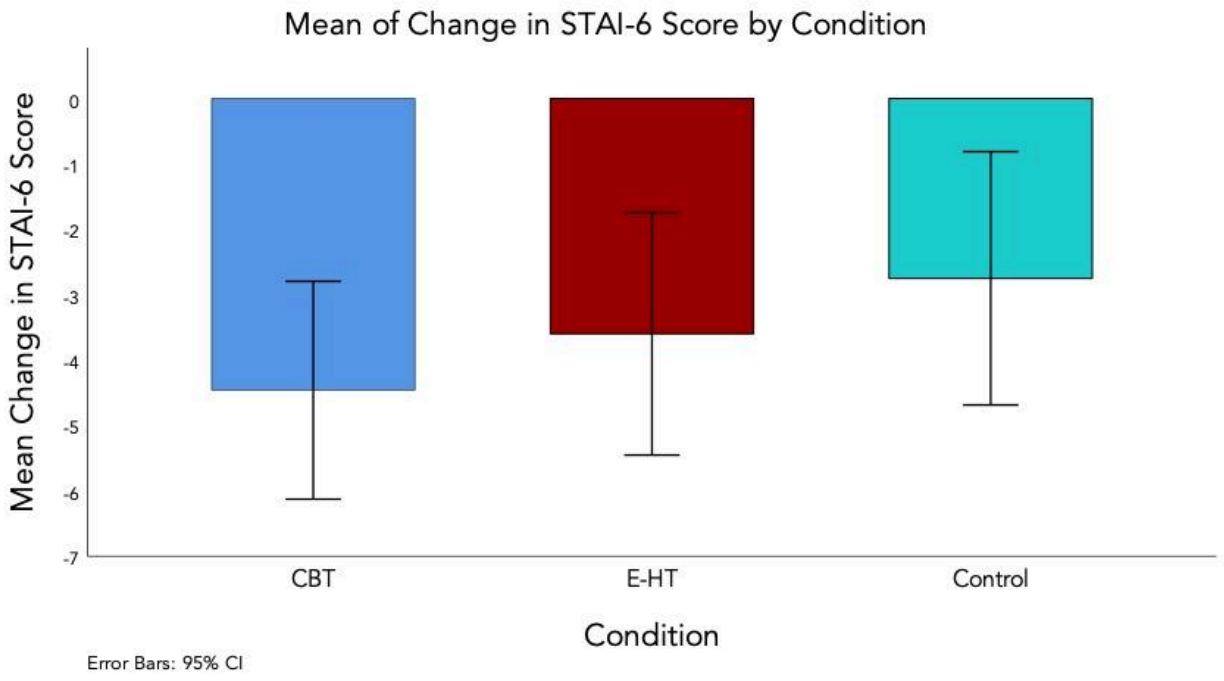
Results indicated that there was no significant effect of condition on change in STAI-6 scores, $F(2, 33) = 1.15$, $p = 0.33$.

Post hoc multiple comparisons indicated that none of the pairwise differences between conditions were statistically significant. Specifically, change scores in the CBT condition did not differ significantly from those in the Existential condition (mean difference = -0.86, $p = 0.75$), nor from the control condition (mean difference = 1.71, $p = 0.30$). Additionally, change scores in the Existential condition did not significantly differ from those in the control condition (mean difference = 0.85, $p = 0.76$).

Thus, hypothesis 3 was not supported, as participants in the CBT and Existential conditions did not show significantly greater reductions in anxiety during the recovery period compared to participants in the control condition (See Figure 3).

Figure 3

Average Change in STAI-6 Scores from Post-TSST to Post-Recovery



Discussion

The purpose of the present study was to examine whether brief, accessible interventions grounded in Cognitive Behavioral Therapy (CBT) or Existential meaning-making could reduce state anxiety in college students following an acute social-evaluative stressor. Using a modified version of the Trier Social Stress Test (TSST), participants completed a public speaking task and were randomly assigned to a CBT-based cognitive reframing worksheet, an existential meaning/values exploration worksheet, or a neutral writing control. Contrary to the study's hypotheses, results indicated no significant differences in state anxiety between conditions immediately following the TSST, after a 30-minute recovery period, or in the magnitude of anxiety reduction during recovery.

Interpretations of Findings

Although previous research has demonstrated the effectiveness of CBT-based strategies and meaning-making interventions in reducing stress and anxiety (Curtiss et al., 2021; Vos & Vitali, 2018), the present findings suggest that very brief, self-guided versions of these interventions may not be sufficient to attenuate acute anxiety following a social-evaluative stressor such as public speaking. The TSST is designed to provoke a robust and reliable stress response through social judgment and uncontrollability (Birkett, 2011), and it is possible that the strength of this stressor outweighed the potential benefits of the brief interventions used in this study.

One explanation for the null findings is that CBT and Existential approaches may require more time, repetition, or therapist guidance to produce measurable effects, particularly when individuals are experiencing heightened physiological arousal. Cognitive restructuring, for example, typically involves practice overal mutliple sessions and feedback to help individuals challenge and modify maladaptive thoughts effectively (Curtiss et al., 2021). Similarly, existential meaning-making may function more as a long-term coping process rather than an immediate anxiety-reduction tool, especially in response to an acute stressor.

Additionally, it is notable that anxiety decreased across all conditions during the recovery period, suggesting that time and disengagement from the stressor alone were sufficient to reduce state anxiety. The structured recovery tasks may have promoted natural emotional regulation by distracting participants and allowing physiological arousal to subside. This pattern is consistent with prior TSST research showing that anxiety typically returns toward baseline levels following a recovery period, regardless of intervention (Hellhammer & Schubert, 2012).

Implications

Despite the lack of condition differences, the findings offer important practical and theoretical implications. First, they suggest that brief, standalone worksheets may not be effective as immediate interventions following acute stress, even when grounded in empirically supported therapeutic frameworks. This has implications for how stress-reduction strategies are implemented in real-world college settings, where students may expect quick solutions to intense stressors such as presentations or exams.

Second, the results highlight the importance of matching interventions to the type and timing of stress. CBT-based and meaning-cenetered strategies may be more effective when used proactively, practiced repeatedly, or applied in anticipation of stress rather than immediately afterward. Colleges may benefit from integrated these approaches into ongoing workshops, counseling services, or academic skills programs rather than relying on one-time interventions.

Finally, the findings reinforce the idea that acute stress responses are highly resilient and normative, particularly in evaluative contexts. Public speaking anxiety is very common among college students (Gallego et al., 2022), and experiencing heightened anxiety during such tasks does not necessarily indicate maladaptive fucntioning. Understanding this may help normalize stress responses and reduce self-criticism among students.

Limitations

Several limitations should be considered when interpreting the results of this study. First, the small sample size likely limited statistical power, reducing the ability to detect small or moderate effects. With only 35 participants divided across three conditions, the study may have been underpowered to observe subtle differences in anxiety reductions.

Second, the interventions were brief, self-guided, and completed only once, which may not reflect how CBT or Existential techniques are typically administered in clinical or applied settings. The lack of therapist guidance or feedback may have reduced participants' engagement or effectiveness in using the strategies.

Third, the study relied exclusively on self-reported state anxiety, which may be influenced by demand characteristics or individual differences in emotional awareness. Although the STAI-6 is a reliable and valid measure, future research could benefit from incorporating physiological indications of stress (e.g., heart rate, cortisol levels) to capture a more comprehensive stress response.

Finally, the use of a modified version of the TSST that only included the speech component may have influenced the strength or duration of the stress response. Including the full TSST protocol (e.g., mental arithmetic) might have produced different patterns of anxiety and recovery.

Future Directions

Future research should examine whether longer or repeated exposure to CBT-based or meaning-making interventions produces stronger effects on stress regulation. Incorporating guided instructions, group discussion, or multiple sessions may increase intervention efficacy.

Additionally, future studied could investigate individual differences that may moderate intervention effectiveness, such as baseline anxiety levels, prior experience with therapy, or person beliefs about stress. Examining these factors may help identify which students are most likely to benefit from specific strategies.

Finally, expanding outcome measures to include behavioral performance or physiological stress markers could provide a more nuanced understanding of how differenct coping strategies influence stress responses beyond subjective anxiety.

Conclusion

In summary, the present study found that brief CBT-based cognitive reframing and Existential meaning-making worksheets did not significantly reduce state anxiety following a public speaking stressor compared to a neutral writing task. While these findings do not support the original hypotheses, the contribute to a growing literature emphasizing the complexity of stress regulation and the limitations of brief interventions in high-stress situations. Understanding when and how therapeutic strategies are most effective remains a critical area for future research aimed at supporting college student mental health.

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